

1. What is AI Automation?

Answer:

AI automation combines artificial intelligence with automated workflows to perform tasks that usually require human intelligence. Unlike traditional automation, AI can analyze data, make decisions, understand language, and improve over time.

Example:

- Traditional automation:
 - If invoice > \$1000 → send to manager.
- AI automation:
 - Read an email.
 - Understand the request.
 - Extract important information.
 - Create a ticket automatically.

2. What is the difference between Automation and AI Automation?

Automation	AI Automation
Rule-based	Learns patterns
Fixed workflow	Dynamic decisions
If-Else logic	Uses AI models
Cannot understand text	Can understand natural language
Deterministic	Context-aware

Example Answer

Traditional automation only follows predefined rules, while AI automation uses machine learning or large language models to interpret information and make intelligent decisions.

3. What AI automation tools have you used?

If you haven't used them professionally, you can answer honestly:

I haven't implemented AI automation in production yet, but I've been learning tools such as:

- n8n
- Zapier
- Make
- OpenAI API
- GitHub Copilot

Because I already work with APIs in React and .NET, integrating AI services fits naturally into my current skill set.

4. What are AI Agents?

Answer

An AI agent is software that can:

- Receive a goal
- Decide what actions to take
- Use tools
- Execute tasks
- Evaluate results

Example:

Customer says:

"Schedule a meeting next Friday."

Agent:

- Reads calendar
- Finds available time
- Books meeting
- Sends invitation

without manual intervention.

5. What is Prompt Engineering?

Answer

Prompt engineering is designing effective instructions for an AI model to generate accurate, relevant, and structured responses.

Example:

Poor prompt:

Summarize this.

Better prompt:

Summarize this email into three bullet points and highlight any action items.

6. Explain an AI workflow.

Example:

Customer Email

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AI reads email

↓

Extract customer name

↓

Determine sentiment

↓

Create CRM ticket

↓

Assign to support

↓

Notify customer

7. What is Retrieval-Augmented Generation (RAG)?

Answer

RAG allows an AI model to retrieve relevant information from an external knowledge base before generating a response. This helps provide more accurate and up-to-date answers without retraining the model.

Example:

- User asks about company policy.
- AI searches the company's documents.
- AI answers using the retrieved information.

8. What are hallucinations in AI?

Answer

Hallucinations occur when an AI generates information that sounds convincing but is inaccurate or unsupported by facts.

To reduce hallucinations:

- Use RAG.
- Provide clear prompts.
- Verify outputs.
- Set appropriate model parameters.
- Include human review for critical tasks.

9. What is an API?

Since you already have API experience, this should be straightforward.

Answer

An API allows different applications to communicate by sending requests and receiving responses.

Example:

React

↓

.NET API

↓

OpenAI API



Response



Display result

10. How would you integrate AI into an existing application?

Sample answer:

I would expose a backend endpoint in .NET that calls the AI service. The React frontend would send user input to the backend, which securely communicates with the AI API, processes the response, and returns only the necessary data to the frontend. This approach protects API keys and centralizes business logic.

11. What challenges exist when using AI?

Answer:

- Hallucinations
- Cost
- Latency
- Privacy
- Data security
- Prompt quality
- Rate limits
- Bias
- Model updates

12. How do you secure AI APIs?

Answer:

- Never expose API keys in frontend code.

- Store keys in environment variables or a secure secrets manager.
 - Use HTTPS.
 - Validate user input.
 - Implement authentication and authorization.
 - Log requests responsibly.
 - Apply rate limiting.
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13. What AI projects would you build?

Examples:

- AI Resume Analyzer
 - AI Customer Support Chatbot
 - AI Document Summarizer
 - AI Meeting Notes Generator
 - AI Code Reviewer
 - AI Email Generator
 - AI SQL Generator
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14. How would AI help a company?

Answer:

- Reduce repetitive work
 - Improve customer support
 - Speed up document processing
 - Generate reports
 - Increase employee productivity
 - Assist developers with coding
 - Analyze customer feedback
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15. Why are you interested in AI Automation?

A good answer tailored to your background:

I've worked mainly with React and .NET building business applications and integrating APIs. I see AI automation as a way to make those applications more intelligent by reducing manual work, improving decision-making, and enhancing user experiences. I'm actively learning AI APIs and workflow automation so I can build solutions that combine traditional software development with AI capabilities.

16. What is JSON

JSON (**JavaScript Object Notation**) is a lightweight text format used to **store and exchange data** between applications. Despite its name, it's not limited to JavaScript—it's widely used with languages like C#, Python, Java, Go, PHP, and many others.

17. HTTP

HTTP request methods (also called **HTTP verbs**) define the action a client wants to perform on a server. If you've built React applications that call a .NET API, you've almost certainly used these.

Method	Purpose	CRUD Operation	Example
GET	Retrieve data	Read	Get all users
POST	Create new data	Create	Add a new user
PUT	Update an entire resource	Update	Replace a user's information
PATCH	Update part of a resource	Update	Change only a user's email
DELETE	Remove data	Delete	Delete a user
HEAD	Get headers only (no response body)	Read	Check if a file exists
OPTIONS	Returns supported HTTP methods	N/A	Browser checks allowed methods (CORS)

18. ASYNC AWAIT

Interview Answer (Memorize This)

async is used to declare a function that performs asynchronous operations. It automatically returns a Promise. **await** is used inside an async function to pause

execution until a Promise is resolved, making asynchronous code easier to read and write than using chained `.then()` calls.

Quick Summary

Keyword	Purpose
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async	Declares an asynchronous function and returns a Promise.
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await	Waits for a Promise to finish before continuing execution.
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Easy way to remember

- **async = "This function works with Promises."**
- **await = "Pause here until the Promise finishes."**

This mental model is enough to explain `async/await` clearly in most JavaScript and React interviews.

Technical Questions

What is an LLM?

A Large Language Model is an AI model trained on large amounts of text to understand and generate human-like language.

What is tokenization?

Tokenization is the process of breaking text into smaller units called tokens, which AI models use as input.

What is temperature?

Temperature controls randomness in AI responses:

- 0.0: More deterministic
- 1.0: More creative

What is embedding?

Embeddings convert text into numerical vectors so systems can compare semantic similarity for tasks like search and retrieval.

What is vector search?

Vector search finds the most semantically similar documents using embeddings instead of exact keyword matching.

Coding Interview Question

How would you call an AI API from React?

A strong answer:

I wouldn't call the AI API directly from React because it would expose the API key. Instead, React would call a secure backend service written in .NET. The backend would authenticate the request, communicate with the AI provider, and return only the required response to the frontend.

Questions worth practicing

- Explain an AI agent.
- Explain RAG.
- Explain embeddings.
- What is vector search?
- What is prompt engineering?
- Explain the AI workflow you've built.
- How would you integrate AI into an existing React/.NET application?
- How do you secure AI APIs?
- How would you automate a business process using AI?
- What are the limitations and risks of using AI?

With your existing React and .NET experience, learning AI integration, prompt engineering, workflow automation (such as n8n), and RAG will position you well for many AI automation developer and automation engineer roles.